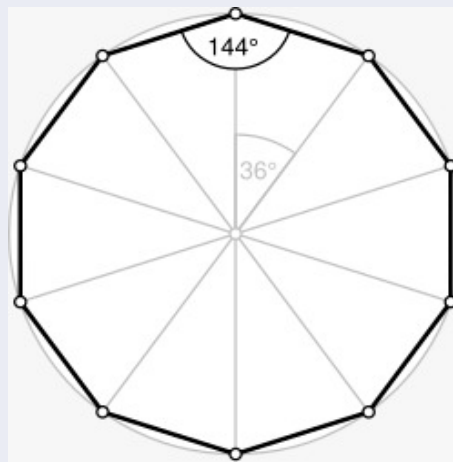


WEEKLY PROBLEM

Decagon Mystery

EXERCISE.

Consider a regular decagon (a 10 sided-polygon) in a circle. All possible diagonals and side of this decagon are drawn.



1. Combinatorics

Calculate the total number of points of intersection of the diagonals that lie **strictly inside** the decagon. Assume that no three diagonals are concurrent (they do not meet at the same point) except at the vertices of the decagon.

2. Geometry & Permutations

If three distinct vertices of the decagon are chosen at random to form a triangle, find the number of such triangles that have **at least one side** in common with the sides of the decagon.

3. Probability

Suppose two distinct diagonals of the decagon are chosen at random. What is the probability that they intersect at a point **inside** the decagon? (Express your answer as a fraction in simplest form).



(Scan this to submit solution!)